

Incidence Perfect NG

User Manual (Beta)



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Incidence Perfect NG is a compact 2-axis inclinometer/incidence meter built around an ESP32-S3 + AMOLED touchscreen. It measures **roll** and **pitch** and provides workflows for **ZERO**, **OFFSET CAL**, **MODE**, **ROTATE**, and **ALIGN**.

This manual describes the current firmware behavior. If something differs on your device, report the **firmware version shown on the splash screen**.

1) Getting Started

Power + Boot

- Connect the device over USB power.
- On boot you'll see the splash screen with the firmware version in the lower-right.
- If the UI was previously set to ROT 180, the splash screen follows that stored orientation.
- If Startup ZERO is enabled, the device begins guided ZERO automatically on cold boot.
- After boot, the live readout screen appears (roll/pitch).

What You're Looking At

- **Top status line** shows:
 - orientation mode (SCREEN UP OR SCREEN VERTICAL)
 - axis view (BOTH, ROLL, PITCH)
 - rotation (ROT 0 OR ROT 180)
 - live state (LIVE OR FROZEN)
 - **Top-right battery label** shows:
 - BAT <percent>% <voltage> V
 - BAT? <percent>% <voltage> v when firmware suspects USB rail without battery pack
 - a steady charge icon when charging is detected
 - **Readouts:**
 - ROLL (left) and PITCH (right) in degrees
 - colors shift for large angles (warning, then critical)
 - **Bottom buttons:**
 - ZERO, AXIS, MODE, ALIGN, ROTATE
-

2) Touch Controls (Everyday Use)

ZERO / OFFSET CAL

Use when the tool is resting in the reference position.

- Tap ZERO.
- ZERO opens a guided workflow:
 - CONFIRM to start stillness + averaging
 - CANCEL to abort
- During apply, a progress bar is shown.
- On completion, values settle around 0.00.
- Long-press ZERO to start guided OFFSET CAL (confirm/cancel workflow).

AXIS

Choose what you want to focus on:

- Tap AXIS to cycle: BOTH -> ROLL -> PITCH -> BOTH.

ROTATE (180 degrees)

Use when the device is physically hard to read and you want to flip the UI.

- Tap ROTATE to toggle ROT 0 / ROT 180.
- Rotation persists after reboot.
- The boot splash follows the same stored rotation.

Freeze / Unfreeze

Use freeze when you want to capture a reading without chasing tiny motion.

- Tap the **readout area** (the roll/pitch values) to toggle LIVE / FROZEN.
- When frozen, the displayed values hold steady.

Touch Layouts: Advanced vs Simple

The device supports two touch layouts:

- **Advanced**: full button row (ZERO, AXIS, MODE, ALIGN, ROTATE)
- **Simple**: two large buttons (ZERO, VIEW)

VIEW in **Simple** combines axis and mode control:

- **Tap VIEW:** cycle axis scope
 - BOTH -> ROLL -> PITCH -> BOTH
- **Medium hold VIEW then release:** toggle MODE
 - SCREEN UP <-> SCREEN VERTICAL
- **Long hold VIEW then release:** switch touch layout
 - Simple <-> Advanced

In **Advanced**:

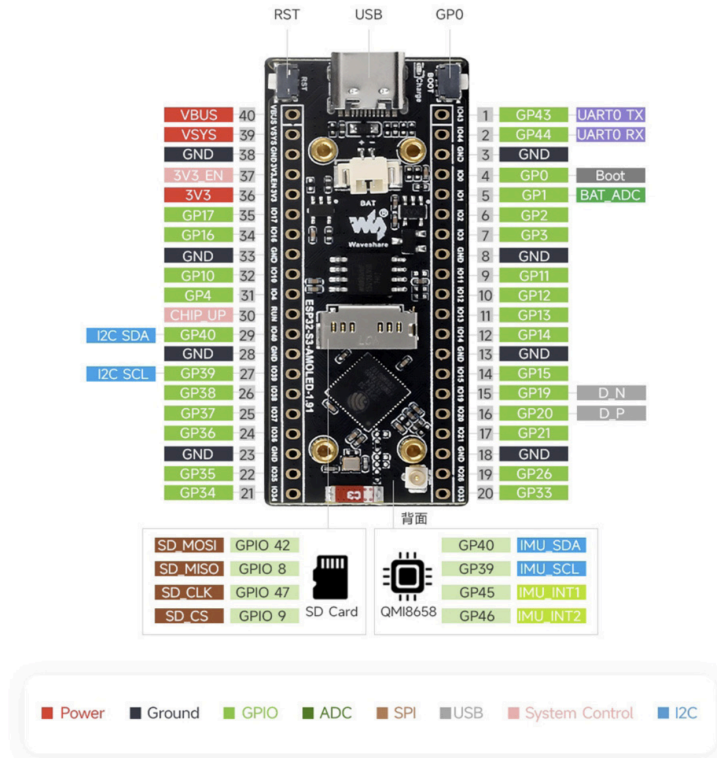
- **Tap MODE:** toggle orientation mode
- **Long hold MODE then release:** switch touch layout to Simple

Tip behavior:

- During hold, the tip line shows what action will happen on release.
 - In Simple, VIEW tip bands are:
 - AXIS (short hold)
 - MODE (medium hold)
 - SWITCH UI (long hold)
-

3) ACTION Button (Physical Control)

The board has a physical **ACTION button** (electrically **GPIO0**, labeled **BOOT/GP0** on the board). It mirrors key actions so you can operate the device when the screen is hard to reach.



In normal measurement mode:

- **Short press:** toggle freeze (LIVE <-> FROZEN)
- **Long press (~1.2s):** cycle AXIS (BOTH -> ROLL -> PITCH)
- **Very long press (~2.2s):** toggle MODE (SCREEN UP <-> SCREEN VERTICAL)
- **Ultra long press (~3.2s):** start guided OFFSET CAL workflow
- **Super long press (~5.0s):** enter deep sleep (press ACTION to wake)

While holding the ACTION button, an on-screen hint shows what will happen on release and a progress indicator for the next threshold. Countdowns are shown as x.x s (with a space before s).

In guided OFFSET CAL workflow:

- **Short press:** CONFIRM
- **Long press (~1.2s):** CANCEL

In guided ZERO workflow:

- **Short press:** CONFIRM

- **Long press (~1.2s):** CANCEL
-

4) MODE (Orientation Change)

MODE changes how the device interprets orientation.

- SCREEN UP: standard "screen facing up" use case
- SCREEN VERTICAL: use case where the tool is used on a vertical surface

Touch Workflow

1. Tap MODE.
2. Orientation switches immediately between SCREEN UP and SCREEN VERTICAL.
3. Mode-specific stored calibration/zero references are loaded automatically.

Serial / Web

- Serial:
 - m: toggle mode
 - u: set SCREEN UP
 - v: set SCREEN VERTICAL
 - Web:
 - MODE toggles immediately
-

5) ALIGN (Mechanical Alignment, 6 Steps)

Use ALIGN after mounting/enclosing the device to remove systematic bias.

This is a guided, 6-orientation capture procedure. The device will prompt you through the positions and ask you to capture each one.

Touch Workflow

1. Tap ALIGN.
2. Follow the on-screen instruction (example: Place tool: SCREEN UP).
3. Tap CAPTURE to record that step.
4. Repeat until all steps are captured.
5. Tap CANCEL to abort safely at any time.
6. After step 6, ALIGN completes immediately and returns to normal operation.

Important:

- ALIGN and OFFSET CAL are separate operations.
- ALIGN does **not** run OFFSET CAL automatically.
- Run guided OFFSET CAL separately in the active MODE when needed.
- If you use both SCREEN UP and SCREEN VERTICAL, run OFFSET CAL in each mode so both mode-specific references are updated.

ACTION Button in ALIGN Workflow

If the screen is hard to access (for example screen-down steps):

- ACTION button short press = CAPTURE
-

6) Serial Control (Optional)

If connected to a PC, you can control the same workflows via serial (115200).

Core commands:

- z: start guided ZERO workflow
- a: AXIS cycle (BOTH -> ROLL -> PITCH)
- r: ROTATE 180 toggle
- c: start ALIGN workflow
- o: start guided OFFSET CAL workflow
- c: context action
 - in ALIGN: CAPTURE
 - in ZERO pending: CONFIRM
 - in OFFSET CAL pending: CONFIRM
 - otherwise: start+confirm guided OFFSET CAL
- y: explicit CONFIRM (ZERO/OFFSET CAL workflows)
- p: explicit CAPTURE (ALIGN only)
- m: toggle mode immediately
- u: set SCREEN UP immediately
- v: set SCREEN VERTICAL immediately
- x or n: cancel active workflows
- d: toggle raw IMU debug stream (5 Hz)
- D: print one raw IMU sample now
- s: print runtime status snapshot
- h or ?: print serial help
- After each serial command response, live scrolling output pauses so you can read feedback.
 - Press Enter, Space, or send g to resume live stream.

Serial and touch workflows are designed to stay synchronized.

7) Web Control (Phone Remote)

Use your phone browser as a remote panel:

1. Connect to the device AP:
 - SSID: IncidencePerfectNG-XXXX
 - xxxx is derived from the board's Wi-Fi MAC identity and should differ between units.
 - Password: incidence-ng
2. Open `http://192.168.4.1`.

If you later switch the device to STA with AP fallback, you can reach it by either:

- its STA IP address
- `http://<hostname>.local`

Important note about hostnames:

- the `.local` suffix is required because hostname access uses mDNS on the local network
- `hostname` by itself is usually not enough
- on some Windows/network setups, `.local` lookup can be inconsistent
- if `hostname.local` does not resolve, use the STA IP address instead

What you get in web UI:

- Live ROLL/PITCH readout with the same status line as the device.
- Battery status (BAT/CHG, percent, and voltage).
- Normal controls: FREEZE, ZERO, OFFSET CAL, AXIS, MODE, ALIGN, ROTATE, SLEEP.
- Context controls only when needed:
 - CANCEL/CONFIRM during guided ZERO and OFFSET CAL
 - CANCEL/CAPTURE during ALIGN
- Progress bar during:
 - ZERO hold/sampling
 - OFFSET CAL hold/sampling
 - ALIGN capture windows
- Expandable diagnostics panel with grouped values:
 - raw sensor vectors
 - mapped vectors
 - corrected vectors

- physics angles + conditioning
- calibration references (bias/zero/align)
- Device settings for:
 - battery indicator behavior
 - Startup ZERO on cold boot
 - readout decimals (1, 2, 3) shared by touch UI and web UI
 - touch input enable/disable
 - optional persistent touch lock across reboot
 - display brightness with a slider (10% to 100%)
- Network settings for:
 - Wi-Fi mode (AP only / STA with AP fallback)
 - hostname
 - STA SSID / password

For practical use, treat hostname access as a convenience and the STA IP as the reliable fallback. - Web UI Appearance settings for: - browser-local Dark / Day theme selection

Surface Displacement Assistant

The web UI includes a setup helper for translating live angle into trailing-edge travel.

Use it to:

- pick a source:
 - ROLL
 - PITCH
 - AUTO (follows the active single-axis view, with preset fallback)
- enter surface depth and units
- choose a preset:
 - Aileron
 - Elevator
 - Flap
 - Rudder
- see live calculated displacement using:
 - $\text{displacement} = \text{depth} * \sin(\text{angle})$

You can also use it as a target-guided setup assistant:

- target by:
 - angle
 - displacement
- set separate UP and DOWN targets
- set tolerance

- optionally set a custom audio approach window
- enable browser-side audio cues while approaching target

Notes:

- all control-surface presets currently default to `PITCH`
- you can still switch source manually to `ROLL` or `AUTO` if your fixture/setup works better that way
- depth and `UP / DOWN` target values are remembered per preset in the browser so you can return to a surface type later
- Audio cues come from the browser device, not the instrument itself.
- On phones/tablets, audio usually requires a tap on `Enable Audio`/arming control first.

Linearity (Experimental)

The web UI also includes an exploratory `Linearity` panel.

Intended workflow:

1. Set your radio/servo to sweep continuously from endpoint to endpoint.
2. Open `Linearity`.
3. Choose source:
 - `ROLL`
 - `PITCH`
 - `Displacement`
4. Press `Record`.
5. Let the sweep run for several cycles.
6. Review the plotted result.

What it shows:

- averaged `UP` and `DOWN` curves
- ideal straight-line reference
- peak deviation from linear response
- hysteresis gap between `UP` and `DOWN`

This feature is currently best treated as a bench-analysis aid/prototype rather than a finalized production workflow.

OTA Update from Web UI

You can update firmware over Wi-Fi from the web page:

1. Open the device web UI (`http://192.168.4.1` in AP mode, or its STA IP/hostname).

2. Expand OTA Update.
3. Select the firmware .bin built for esp32s3.
4. Enter the exact target firmware version shown in the release notes (YYYY.M.X).
5. Provide matching SHA-256:
 - use browser auto-calc if available, or
 - paste SHA-256 from release notes / local hash command below.
6. Leave Force unchecked for normal updates.
7. Tap Upload & Install.
8. Wait for reboot, reconnect, and verify the splash-screen firmware version changed to the target version.
9. If upload is rejected, correct version/SHA and retry (device keeps running current firmware when rejected).

If browser SHA-256 auto-calc is unavailable, compute hash locally and paste it:

```
Get-FileHash .\.\.pio\build\esp32s3\firmware.bin -Algorithm SHA256 | Select-Object -
ExpandProperty Hash
```

If you build firmware from source, the default artifact path is:

- .pio/build/esp32s3/firmware.bin

Important:

- OTA updates only the firmware app image.
- If bootloader or partition layout changes, update once over USB first.
- OTA safety gates reject packages when:
 - target version is not newer than current version (unless force is enabled),
 - uploaded file hash does not match the provided SHA-256.
- For release testing, always verify both:
 - positive path: valid newer version + matching SHA succeeds,
 - negative path: same-version (without force) or wrong SHA is rejected.

Network Recovery

- In the web Network panel, use Recover AP Mode to force AP-only and clear saved STA credentials.
- Physical fallback:
 - reboot device, then immediately press+hold ACTION (GPIO0) continuously through startup for ~2 s
 - do not hold ACTION before power-on/reset, because GPIO0 is a boot-strap pin
 - this applies the same AP recovery defaults (AP-only + cleared STA credentials)

Battery note:

- In web Device Settings, Battery can be set to:
 - Auto detect (heuristic/default)
 - Installed (force battery-present UI semantics)
 - No battery installed (hide battery indicators in both web and touch UI)
- In the same Device Settings, Startup ZERO can be set to:
 - Enabled (default; guided ZERO starts automatically on cold boot)
 - Disabled (keep the saved zero reference until you explicitly run ZERO)
- In Device Settings, Readout decimals can be set to 1, 2, or 3.
- In Device Settings, Touch input can be disabled if you want to tape the device to a surface without accidental taps.
 - If Persist touch lock across reboot is left off, reboot restores touch automatically.
 - If persistence is enabled, touch stays disabled until re-enabled from a non-touch path such as the web UI or ACTION button workflow.
- In Device Settings, Brightness controls the physical display brightness from 10% to 100%.
- Current charging state is inferred from battery-voltage trend in firmware.
- The web/API exposes this via `battery_charging_inferred`.
- Battery-pack presence is best-effort on this hardware; when firmware suspects "USB powered, no pack", web/API reports `battery_present=false` with `battery_present_inferred=true`.
- On some board/charger combinations, no-pack detection may stay at BAT/CHG even with battery disconnected (shared rail behavior).

Web Refresh Behavior (Current)

- Live angles refresh about every 120 ms.
- Workflow state and diagnostics refresh about every 600 ms.
- Network status refreshes every 3000 ms.

Important:

- Web, touch, serial, and ACTION button all share the same workflows and states.
- If one interface starts a workflow, the other interfaces should reflect it.

8) Troubleshooting

Touch Feels Hard To Trigger

- Use ACTION button alternatives for critical actions.
- Try deliberate taps (not swipes) centered on the button.

Serial Monitor Doesn't Resume After Reset

- Some setups require closing/reopening the serial monitor after reset.
- Always report the firmware version shown on splash if you see inconsistent serial behavior.

MODE Doesn't Apply

- MODE is immediate in current firmware.
 - If orientation does not change, verify command source:
 - touch MODE,
 - serial m/u/v,
 - ACTION very-long press.
-

9) Measurement Math (How It Works)

This device estimates orientation from gravity measured by the accelerometer and short-term rate from the gyro.

Accelerometer Tilt Angles

In the tool frame (a_x , a_y , a_z), the firmware computes:

```
roll_acc = atan2(a_y, a_z)
pitch_acc = atan2(-a_x, sqrt(a_y^2 + a_z^2))
```

Then values are converted to degrees.

Why pitch uses $\sqrt{a_y^2 + a_z^2}$:

- It uses the gravity component orthogonal to the pitch axis.
- This reduces cross-coupling from roll.
- It stays better conditioned as pitch approaches large angles.

Sensor Fusion

The live angles are fused with a complementary filter:

- Gyro integration gives smooth short-term motion response.
- Accelerometer tilt gives long-term absolute reference to gravity.

This combination is stable and responsive for field use.

Why Roll Becomes Weak Near +/-90 deg Pitch

Near pitch +/-90 deg, gravity aligns mostly with one axis, so roll information in the other components becomes very small. At that geometry, roll is effectively underdetermined from gravity alone.

Current firmware behavior:

- Above about $|\text{pitch}| > 80$ deg, roll is intentionally attenuated to avoid unstable swings.

Practical Guidance For Large-Angle Work

For tasks near vertical (for example checking a table-saw blade relative to the table):

1. Use the orientation/mode that makes your primary measurement axis map to PITCH.
2. Switch AXIS to PITCH for focus.
3. Run OFFSET CAL and ZERO in the same physical setup/orientation you will measure in.
4. Treat ROLL near +/-90 deg as secondary information.

10) Validation + Feedback

For validation and release readiness, use:

- ../testing/hardware-validation-checklist.md
- ../testing/validation-session-template.md

When reporting an issue, include:

1. Firmware version (from splash)
2. Exact steps to reproduce
3. Expected vs actual result
4. Photos/video if UI-related

Appendix A: Hardware Notes

For reference bring-up settings (Arduino IDE), see:

- ../hardware/board-settings.md
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Authorship

This firmware and UI were developed by **Per Takman**, with assistance from **OpenAI Codex (Codex CLI)**.

License And Warranty

- The project source code is released under the MIT License.
- The software is provided "AS IS", without warranty of any kind.
- Third-party manuals, schematics, and library dependencies remain under their respective original licenses/terms.